

Course Information	Details
Course Code	CSE 3226
Course Title	Mathematical Analysis for Computer Science
Credit Hours	3.00 (Theory)
Pre-requisite Courses	CSE 2111, MATH 2305
Class Structure	2 Classes per Week (80 Minutes per Session)
Core Reference Books	1. Sheldon M. Ross : Introduction to probability Models 2. Ronald L. Graham, Donald E. Knuth, Oren Patashnik : Concrete Mathematics

Evaluation Component	Assessment Strategy	Marks
Attendance	Regular attendance throughout the semester	10
Class Test 1 (CT1)	Written assessment	15
Class Test 2 (CT2)	Written assessment	15
Continuous Assessment	Best score from CT1 & CT2 will be counted	15
Assignment	Assignment	5
Midterm Examination	MCQ & Written Examination	30
Final Examination	MCQ & Comprehensive Written Examination	40
Total		100

Evaluation Component	Assessment Strategy	Marks
Week / Date	Lecture / Event	Topic / Description
Week 1 (06 Jun – 12 Jun)	Lecture 01	Introduction to Mathematical Analysis for Computer Science and Review of Discrete Structures
	Lecture 02	Recurrent Problems and Formation of Recurrence Relations
Week 2 (13 Jun – 19 Jun)	Lecture 03	Solving Linear Recurrence Relations and Applications
	Lecture 04	Generating Functions and Their Computational Applications
Week 3 (20 Jun – 26 Jun)	Lecture 05	Advanced Generating Functions and Sequence Analysis
	Lecture 06	Mathematical Sums, Summation Techniques, and Asymptotic Behavior
	Holiday	Ashura (26 June 2026) – No Class
Week 4 (27 Jun – 03 Jul)	Lecture 07	Integer Functions: Floor, Ceiling, and Modular Arithmetic
	Lecture 08	Foundations of Number Theory and Divisibility
	CT-1	Class Test 1 (CT1) covering Lectures 01–08
Week 5 (04 Jul – 10 Jul)	Lecture 09	Prime Numbers, GCD, Euclidean Algorithm, and Modular Applications
	Lecture 10	Binomial Theorem and Combinatorial Applications
Midterm Period (13 Jul – 24 Jul)	Midterm Examination	Midterm Examinations (No Regular Classes)
Week 6 (25 Jul – 31 Jul)	Lecture 11	Midterm Script Discussion and Introduction to Probability Theory
	Lecture 12	Probability Distributions and Expectation

Evaluation Component	Assessment Strategy	Marks
Week 7 (01 Aug – 07 Aug)	Lecture 13	Random Variables and Expected Value Applications
	Holiday	July Revolution Day (05 August 2026) – No Class
	Lecture 14	Introduction to Stochastic Processes
Week 8 (08 Aug – 14 Aug)	Lecture 15	Discrete-Time Markov Chains and State Transition Analysis
	Lecture 16	Continuous-Time Markov Chains and Applications
Week 9 (15 Aug – 21 Aug)	Lecture 17	Birth-Death Processes and Queue Formation Models
	Lecture 18	Queuing Theory Fundamentals and M/M/1 Model
Week 10 (22 Aug – 28 Aug)	Lecture 19	Class Test 2 (CT2) covering Lectures 11–18
	Holiday	Eid-e-Miladun Nabi (26 August 2026) – No Class
	Lecture 20	Multi-Server Queues: M/M/C Model
Week 11 (29 Aug – 03 Sep)	Lecture 21	M/G/1 and M/D/1 Queue Models
	Lecture 22	G/M/1 Model and Solution of Network Queues
Week 12 (04 Sep – 08 Sep)	Holiday	Shuvo Janmashtami (04 September 2026) – No Class
	Presentation Assessment	Mathematical Modeling, Queuing Analysis, and Stochastic Process Presentation & Course Review
Final Examination Period (09 Sep – 21 Sep)	Final Examination	Comprehensive Final Examination Window